

Counterexamples to EFX for Submodular and Subadditive Valuations

Simon Mackenzie*

Mashbat Suzuki†

Abstract

The existence of EFX allocations is a fundamental question in fair division. In this paper, we construct a three-agent, eight-good instance with monotone subadditive valuations such that no allocation satisfies α -EFX for any $\alpha > \frac{1}{\sqrt[3]{2}} \approx 0.89$. We also provide a closely related three-agent, eight-good instance with submodular (in fact weighted coverage) valuations for which no EFX allocation exists.

A key feature of our construction is its symmetry: the agents' valuations are identical up to a relabeling of the goods. Thus, EFX can fail even when agents differ only in how the goods are labeled. This symmetry makes the counterexamples compact and human-verifiable, yielding simple combinatorial obstructions to the existence of EFX.

1 Introduction

A mathematical study of fairness in resource-allocation settings can be traced back to the work of [Steinhaus \[1948\]](#). Over the past several decades, many fairness notions have been proposed to capture different desiderata across a range of application domains. Among these, envy-freeness (EF) has emerged as a central notion: an allocation is envy-free if no agent prefers another agent's bundle to their own.

In the setting of indivisible goods, it is impossible to guarantee envy-freeness. This is evident even with just two agents and a single good: the agent who does not receive the good envies the one who does. As a result, the discrete fair division literature has developed meaningful relaxations of envy-freeness. Among these, one of the strongest and arguably closest analogues of envy-freeness is *envy-freeness up to any item* (EFX), proposed by [Caragiannis et al. \[2016\]](#). An allocation is EFX if, whenever an agent envies another, this envy can be eliminated by removing *any* good from the envied agent's bundle. Whether EFX allocations always exist has been one of the central questions in fair division [Procaccia \[2020\]](#).

Recently, [Akrami et al. \[2026\]](#) constructed an instance with general monotone valuations that admits no EFX allocation. However, their counterexample was obtained through a large SAT-based search and is not readily verifiable by hand. Moreover, the instance is described by three large tables specifying the ranking of every subset, amounting to hundreds of entries. This makes it difficult to identify structural features of the construction or to isolate the underlying combinatorial obstacles to the existence of EFX allocations.

This raises a natural question: can EFX fail in familiar, well-studied, and structured valuation classes, and can the reason for this failure be made transparent and human-verifiable? We answer both questions affirmatively.

*UNSW Sydney; simon.william.mackenzie@gmail.com

†UNSW Sydney; mashbat.suzuki@unsw.edu.au

Our first result shows that the non-existence of EFX is not confined to general monotone valuations, but already arises in a highly structured valuation class of weighted coverage functions (a subclass of submodular functions).

Theorem 1.1. *There exists a fair division instance with three agents and eight goods, where every agent has a **weighted coverage valuation**, such that **no allocation is EFX**. In particular, EFX need not exist for **monotone submodular** valuations.*

Our second result concerns approximate EFX. For a fixed constant $\alpha \in [0, 1]$, an allocation is α -EFX if each agent's value for her own bundle is at least an α -fraction of the value she assigns to any other agent's bundle **after the removal of any one good from that bundle**. We construct an instance with monotone subadditive valuations such that no allocation is α -EFX for any $\alpha \in (2^{-1/6}, 1]$, where $2^{-1/6} \approx 0.891$. To the best of our knowledge, this is the first upper bound strictly below 1 on the best possible approximation factor for EFX under monotone subadditive valuations. Since $\frac{1}{2}$ -EFX allocations are known to exist for subadditive valuations, our result places the optimal guarantee for this class between $1/2$ and $2^{-1/6}$.

Theorem 1.2. *There exists a fair division instance with three agents and eight goods, in which every agent has a monotone subadditive valuation, such that no allocation is α -EFX for any $\alpha \in (2^{-1/6}, 1]$.*

Both of our results exhibit a high degree of symmetry. In particular, the agents' valuations are identical up to a **relabeling of the goods**. More concretely, the eight goods are grouped into types as

$$A, A, B, B, C, C, x, y.$$

Agents make **no distinction** between goods of the same type. We first describe the preferences of the first agent succinctly: we assign values to all singletons and pairs, list a small number of exceptional triples, and then extend the preferences to all remaining bundles by assigning each such bundle the value of its **highest-valued triple**.

The valuations of the second and third agents are obtained by applying the same rule as for the first agent, but with the item types A, B, C cyclically rotated. Thus, what the first agent regards as an **A-good**, the second agent regards as a B -good, and the third agent regards as a C -good, while the special goods x and y keep their names. In this sense, all agents share the same valuation structure; they differ only in how the goods are named.

Note that the instances we construct use the same number of agents ($n = 3$) and number of items ($m = 8$) as the example given by Akrami et al. [2026]. Thus these structured counterexamples do not require more goods than the general monotone non-existence construction.

1.1 Related Work

Early work on EFX and near-envy-freeness for indivisible goods includes Gourvès et al. [2014] and Caragiannis et al. [2019]. Plaut and Roughgarden [2020] studied EFX for general monotone valuations and proved several foundational positive results, including existence for identical valuations. This makes the symmetry in our examples noteworthy: our agents are **not identical**, but they are all obtained from **one valuation by cyclically relabelling the goods**.

For three agents, the positive theory goes beyond additive valuations: Akrami et al. [2022] prove that EFX exists whenever two agents have arbitrary monotone valuations and the remaining agent has an **MMS-feasible valuation**. Our counterexamples also have three agents, so they necessarily lie outside this positive regime. For subadditive valuations, general exact EFX remains much less understood. A $1/2$ -EFX allocation is known to exist for subadditive valuations; see

Plaut and Roughgarden [2020] and the discussion in Chaudhury et al. [2021]. Barman and Suzuki [2026] show that under **subadditive** valuations, one can achieve 1/2-EFX, EF1 and two approximation to optimal Nash Social Welfare simultaneously. Our subadditive construction shows that no universal guarantee can exceed $2^{-1/6} \approx 0.891$. Thus the best possible universal approximation factor for monotone subadditive valuations lies between 1/2 and $2^{-1/6}$.

The present work was motivated by the counterexample of Akrami et al. [2026], but the instances here are different. Their paper gives a SAT-generated monotone counterexample with three agents and eight goods, and extends the construction to all $n \geq 3$ and $m \geq n + 5$; their work also shows that this item count is minimal for the monotone non-existence phenomenon. Their basic three-agent example is specified by three full linear orders over all 2^8 bundles. In contrast, our agents are cyclic relabellings of one typed valuation with only a small number of rank levels. Thus our construction uses the same number of agents and goods, and was motivated by the same question, but it is not the same ordinal instance. We instead construct a compact cyclic typed obstruction and realise it in much more structured valuation classes, with a human-checkable case analysis.

On the submodular side, positive EFX results are known in binary regimes. For example, Babaioff et al. [2021] prove EFX existence for submodular valuations with dichotomous marginals, and Bu et al. [2023] extend existence to general binary valuations. Our weighted-coverage example shows that exact EFX can already fail in a standard and widely used subclass of monotone submodular valuations once one leaves these binary regimes. It also escapes the known three-agent MMS-feasible positive theorem.

2 A Cyclic Ordinal Obstruction to EFX

We first construct a monotone ordinal counterexample. The agents have weak preferences over bundles, represented by monotone rank functions. We define EFX directly in terms of these weak preferences, so this formulation is purely ordinal and does not rely on **cardinal valuations**. Let M be a set of indivisible goods. A *weak preference order* for agent i is a complete and transitive relation $\succeq_i$ over the set of all subsets of items 2^M .

The induced indifference relation $\sim_i$ is defined by

$$S \sim_i T \iff S \succeq_i T \text{ and } T \succeq_i S.$$

The equivalence classes of $\sim_i$ are called the *rank classes* of agent i . A function

$$r_i : 2^M \rightarrow \mathbb{Z}_{\geq 0}$$

is a *rank representation* of $\succeq_i$ if, for every $S, T \subseteq M$,

$$S \succeq_i T \iff r_i(S) \geq r_i(T).$$

Thus, bundles in the same indifference class receive the same rank, while strictly preferred bundles receive strictly larger ranks. We set $r_i(\emptyset) = 0$. The preferences we construct are monotone: for every agent i , if $S \supseteq T$, then $S \succeq_i T$, and hence $r_i(S) \geq r_i(T)$.

Under the weak preferences an allocation $X = (X_1, \dots, X_n)$ is EFX if, for every pair of agents $i, j \in N$ and every good $g \in X_j$, $X_i \succeq_i X_j \setminus \{g\}$. Equivalently, using the induced rank function,

$$r_i(X_i) \geq r_i(X_j \setminus \{g\}) \quad \text{for every } i, j \in N \text{ and every } g \in X_j.$$

It will be useful to isolate the single-agent version of the EFX condition. Given an allocation $X = (X_1, \dots, X_n)$, we say that X is *EFX-feasible for agent i* if

$$X_i \succeq_i X_j \setminus \{g\} \quad \text{for every } j \in N \text{ and every } g \in X_j.$$

We say that agent i *strongly-envies* agent j if there exists $g \in X_j$ such that

$$X_i \prec_i X_j \setminus \{g\}.$$

An allocation is EFX if and only if it is EFX-feasible for every agent.

We now state the main result of this section.

Theorem 2.1. *There exists an instance with three agents and eight items, with monotone weak preference profile $(\succeq_0, \succeq_1, \succeq_2)$, such that no allocation satisfies EFX.*

In Section 2.1, we describe the instance used to prove the theorem. In Section 2.2, we prove that no allocation in this instance satisfies EFX.

2.1 Instance Description

The instance consists of three agents $N = \{0, 1, 2\}$ and eight goods $M = \{0, \dots, 7\}$.

The construction is generated from a single base weak order $\succeq_0$, which represents the preferences of agent 0. The preferences of the other two agents are obtained by cyclically relabelling the goods. Let

$$\sigma = (012)(345)$$

be a permutation of M , written in cycle notation. We define the preferences $\succeq_1$ and $\succeq_2$ of agents 1 and 2 by

$$S \succeq_1 T \iff \sigma(S) \succeq_0 \sigma(T), \quad \forall S, T \subseteq M, \quad (1)$$

and

$$S \succeq_2 T \iff \sigma^2(S) \succeq_0 \sigma^2(T), \quad \forall S, T \subseteq M. \quad (2)$$

Here, $\sigma(S) = \{\sigma(g) : g \in S\}$ and $\sigma^2(S) = \sigma(\sigma(S))$. Since σ^3 is the identity permutation, we have $\sigma^3(S) = S$ for every $S \subseteq M$.

Example 2.2. Suppose that under the base weak order of agent 0 we have

$$\{0, 3\} \succeq_0 \{1, 6\}.$$

Since

$$\sigma(\{2, 5\}) = \{0, 3\} \quad \text{and} \quad \sigma(\{0, 6\}) = \{1, 6\},$$

the corresponding comparison for agent 1 is

$$\{2, 5\} \succeq_1 \{0, 6\}.$$

Similarly, since

$$\sigma^2(\{1, 4\}) = \{0, 3\} \quad \text{and} \quad \sigma^2(\{2, 6\}) = \{1, 6\},$$

the corresponding comparison for agent 2 is

$$\{1, 4\} \succeq_2 \{2, 6\}.$$

Thus, the three agents have identical weak preference structures up to a cyclic relabelling of the goods. Hence, the preferences of every agent are fully specified once the base weak order $\succeq_0$ is given.

Base weak order $\succeq_0$. To define the weak preference order $\succeq_0$, it suffices to specify the corresponding rank function $r_0 : 2^M \rightarrow \{0, 1, 2, \dots, 7\}$. We set $r_0(\emptyset) = 0$. Partition the first six goods into three types:

$$A = \{0, 3\}, \quad B = \{1, 4\}, \quad C = \{2, 5\};$$

and let the remaining two items denoted by $x = 6$ and $y = 7$.

We now define $r_0(S)$ depending on the cardinality of S .

- For singletons, set

$$r_0(\{g\}) = 1 \quad \text{for every } g \in M.$$

- For $|S| = 2$, the rank depends only on the types of the two goods, as shown in Table 1. For example, any pair with one good from A and one good from B has rank 2, whereas any pair with one good from C and the good x has rank 1. Diagonal entries apply to pairs consisting of two distinct goods of the same type.

Table 1: Ranks r_0 of sets with cardinality two.

	A	B	C	x	y
A	1	2	2	4	6
B	2	1	5	1	3
C	2	5	1	1	3
x	4	1	1	–	1
y	6	3	3	1	–

- For $|S| = 3$, we call S *exceptional* if it contains one good from $A \cup \{x\}$, one good from B , and one good from C . Exceptional triples are assigned rank 7. Every other triple inherits the largest rank among its three internal pairs:

$$r_0(S) = \begin{cases} 7, & \text{if } S \text{ is exceptional,} \\ \max_{\{g,h\} \subseteq S} r_0(\{g,h\}), & \text{otherwise.} \end{cases}$$

The exceptional triples are listed in Table 2.

Table 2: **Exceptional triples**. These triples have the highest possible rank of 7.

Type	Triples
ABC	012, 015, 024, 045, 123, 135, 234, 345
BCx	126, 156, 246, 456

- For $|S| \geq 4$, set

$$r_0(S) = \max_{\substack{T \subseteq S \\ |T|=3}} r_0(T).$$

Equivalently, a set of size at least four has rank 7 exactly when it contains an exceptional triple; otherwise, it inherits the largest rank of an internal pair.

For every subset $S \subseteq M$, we set

$$r_1(S) = r_0(\sigma(S)) \quad \text{and} \quad r_2(S) = r_0(\sigma^2(S)). \quad (3)$$

By Equations (1) and (2), the functions r_1 and r_2 are valid rank representations of $\succeq_1$ and $\succeq_2$, respectively.

Observation 2.3. For each agent $i \in N$, the rank function r_i is monotone.

Proof. It suffices to prove the claim for r_0 , since r_1 and r_2 are obtained from r_0 by relabelling. Consider any two sets of items $S \subsetneq T$. If $S = \emptyset$, the claim is immediate. If $|S| = 1$, then $r_0(S) = 1$, since every non-empty set has rank at least 1, we also have $r_0(T) \geq 1$.

Now suppose that $|S| = 2$, and choose $h \in T \setminus S$. Then $U = S \cup \{h\}$ is a triple contained in T . If U is exceptional, then $r_0(U) = 7 \geq r_0(S)$. Otherwise,

$$r_0(U) = \max_{\{a,b\} \subseteq U} r_0(\{a,b\}) \geq r_0(S).$$

If $|T| = 3$, then $T = U$. If $|T| \geq 4$, then by definition $r_0(T)$ is the maximum rank of an internal triple and hence $r_0(T) \geq r_0(U)$. Therefore $r_0(T) \geq r_0(S)$.

Finally, suppose that $|S| \geq 3$. If S contains an exceptional triple, then so does T . Otherwise, every internal pair of S is also an internal pair of T . In either case, we have $r_0(T) \geq r_0(S)$. $\square$

2.2 Proof of Non-Existence of EFX Allocations

Lemma 2.4 (Cyclic Symmetry). *If an allocation $X = (X_0, X_1, X_2)$ is EFX, then the allocation*

$$X^\sigma = (\sigma(X_1), \sigma(X_2), \sigma(X_0))$$

is also EFX.

Proof. For every bundle $S \subseteq M$, the definitions of the preferences and rank functions imply

$$r_{i+1}(S) = r_i(\sigma(S)),$$

where indices are read modulo 3. Indeed, this follows from the identity $r_i(S) = r_0(\sigma^i(S))$, together with the fact that $\sigma^3 = \text{Id}$.

Fix agents $i, j \in N$ and an arbitrary good $g \in X_j^\sigma$. Since $X_j^\sigma = \sigma(X_{j+1})$, there exists $h \in X_{j+1}$ such that $g = \sigma(h)$, and thus $X_j^\sigma \setminus \{g\} = \sigma(X_{j+1} \setminus \{h\})$. Because X is EFX we have that,

$$r_{i+1}(X_{i+1}) \geq r_{i+1}(X_{j+1} \setminus \{h\}).$$

Using $r_{i+1}(S) = r_i(\sigma(S))$, we get

$$r_i(X_i^\sigma) \geq r_i(X_j^\sigma \setminus \{g\}).$$

Since i, j , and $g \in X_j^\sigma$ were arbitrary, we have that indeed X^σ is EFX. $\square$

Note that Lemma 2.4 allows us to rotate size patterns. In particular, if there exists an EFX allocation with size pattern $(|X_0|, |X_1|, |X_2|)$, then there also exist EFX allocations with size patterns

$$(|X_1|, |X_2|, |X_0|) \quad \text{and} \quad (|X_2|, |X_0|, |X_1|).$$

This is because cyclic relabelling preserves bundle sizes; that is, $|\sigma(S)| = |S|$ for each $S \subseteq M$.

Since there are eight items and three agents, this rotation of size patterns allows us to restrict attention to allocations of the following three types:

- $|X_0| \leq 1$;
- $(|X_0|, |X_1|, |X_2|) = (2, 2, 4)$;
- $(|X_0|, |X_1|, |X_2|) = (2, 3, 3)$.

We will show that no allocation of any of these three types satisfies EFX, thereby proving Theorem 2.1.

Proposition 2.5. *Every allocation $X = (X_0, X_1, X_2)$ with $|X_0| \leq 1$ fails to satisfy EFX.*

Proof. Consider first the case $|X_0| = 0$. Then agent 0's bundle has rank 0, while some other agent receives at least two items. After deleting one item from that other bundle, the remaining bundle is non-empty and has positive rank. Hence agent 0 strongly envies that agent.

Now suppose $|X_0| = 1$, which implies that $r_0(X_0) = 1$. Since the other two agents receive seven goods between them, one of them receives a bundle F of size at least four.

We claim that there is some $g \in F$ such that $F \setminus \{g\}$ has rank at least 2. To see this, observe that pair types of rank at least 2 are

$$AB, \quad AC, \quad Ax, \quad Ay, \quad BC, \quad By, \quad Cy,$$

Thus it suffices to show that F cannot avoid all of these pair types. Indeed, if $y \in F$, then y can only be paired with x , so $|F| \leq 2$. If $x \in F$ and $y \notin F$, then x cannot be paired with an A -good, and avoiding BC leaves at most x together with the two B -goods or the two C -goods; again $|F| < 4$. Finally, if neither x nor y belongs to F , then $F \subseteq A \cup B \cup C$. Avoiding AB , AC , and BC then allows goods from only one of the three types A, B, C , giving at most two goods.

Therefore, F contains a pair of rank at least 2. Since $|F| \geq 4$, there is some $g \in F$ whose removal leaves this pair in $F \setminus \{g\}$. Hence $r_0(F \setminus \{g\}) \geq 2$, while $r_0(X_0) = 1$. Thus agent 0 strongly envies the recipient of F after the deletion of g . Therefore X cannot be EFX. $\square$

Reference tables for the case analysis. The following tables are included only to make the subsequent checks easier to verify. They are obtained directly from the base rank function r_0 and the identities

$$r_1(S) = r_0(\sigma(S)), \quad r_2(S) = r_0(\sigma^2(S)).$$

On item types, the cyclic relabelling sends

$$A \mapsto B \mapsto C \mapsto A, \quad x \mapsto x, \quad y \mapsto y.$$

Table 3: Pair ranks for all three agents.

(a) r_0 -ranks						(b) r_1 -ranks						(c) r_2 -ranks					
	A	B	C	x	y		A	B	C	x	y		A	B	C	x	y
A	1	2	2	4	6	A	1	5	2	1	3	A	1	2	5	1	3
B	2	1	5	1	3	B	5	1	2	1	3	B	2	1	2	4	6
C	2	5	1	1	3	C	2	2	1	4	6	C	5	2	1	1	3
x	4	1	1	–	1	x	1	1	4	–	1	x	1	4	1	–	1
y	6	3	3	1	–	y	3	3	6	1	–	y	3	6	3	1	–

Table 4: Exceptional triples for all three agents.

Agent	Type	Exceptional triples
0, 1, 2	ABC	012, 015, 024, 045, 123, 135, 234, 345
0	BCx	126, 156, 246, 456
1	ABx	016, 046, 136, 346
2	ACx	026, 056, 236, 356

In the case analysis, expressions such as Ay , $BBCC$, or $CCx \mid AAB$ denote type multisets. For example, Ay denotes any pair consisting of one A -good and y . Since the ranks are defined by the type table and cyclic relabelling, the identities of the two copies inside each of A, B, C are irrelevant.

Lemma 2.6 (First-pair restriction). *Let $X = (X_0, X_1, X_2)$ be an allocation such that*

$$|X_0| = 2 \quad \text{and} \quad |X_1|, |X_2| \geq 2.$$

If agent 0 is EFX-feasible, then X_0 has one of the following types:

$$Ax, \quad Ay, \quad BC, \quad By, \quad Cy.$$

Proof. It suffices to show that, for agent 0 to be EFX-feasible, we must have $r_0(X_0) \geq 3$. This proves the lemma, since the pair types of rank at least 3 are exactly Ax, Ay, BC, By, Cy .

There are two possible cases for the sizes of the other two bundles: either $|X_1| = |X_2| = 3$, or one of them has size four.

First suppose that $|X_1| = |X_2| = 3$. If $y \notin X_0$, then y belongs to one of the two triples. This triple contains y together with at least one good other than x . Hence, by deleting a good, we can always obtain a pair of type Ay, By , or Cy , each of which has rank at least 3. Since agent 0 is EFX-feasible, it follows that $r_0(X_0) \geq 3$.

Now suppose that $y \in X_0$. The only way to have $r_0(X_0) < 3$ is $X_0 = xy$. In that case, the remaining six goods are precisely those in $A \cup B \cup C$. However, these six goods cannot be partitioned into two triples whose internal pairs all have rank at most 1: every such triple contains a pair of type AB, AC , or BC . Hence $X_0 \neq xy$, and therefore $r_0(X_0) \geq 3$.

It remains to consider the case in which one of X_1, X_2 is a size four-bundle; denote it by F . Suppose, for a contradiction, that $r_0(X_0) \leq 2$. Then every pair contained in F must have rank at most 2, since each such pair is contained in some deletion of F . Thus F cannot contain any pair of type

$$Ax, \quad Ay, \quad BC, \quad By, \quad Cy.$$

The only size four sets avoiding all these pairs are $AABB$ and $AACC$.

In either case, some deletion of F has rank 2, so EFX-feasibility forces $r_0(X_0) = 2$. If F has type $AABB$, then the goods outside F have types C, C, x, y ; among pairs of rank at most 2, none has rank 2. If F has type $AACC$, then the goods outside F have types B, B, x, y , and again no pair of rank at most 2 has rank 2. In both cases, this contradicts $r_0(X_0) = 2$. Therefore $r_0(X_0) \geq 3$, completing the proof. $\square$

Allocations of type $(2, 2, 4)$

Proposition 2.7. *No allocation of size pattern $(2, 2, 4)$ is EFX.*

Proof. Suppose, for the sake of contradiction, that there exists an EFX allocation $X = (X_0, X_1, X_2)$ with size pattern $(|X_0|, |X_1|, |X_2|) = (2, 2, 4)$. Since X is EFX, agent 0 is EFX-feasible. Hence every deletion of X_2 has rank at most $r_0(X_0)$:

$$r_0(X_2 \setminus \{g\}) \leq r_0(X_0) \quad \text{for every } g \in X_2.$$

In particular, X_2 contains no exceptional triple, since exceptional triples have rank 7, while X_0 is a pair and hence has rank at most 6. Also, every pair contained in X_2 has rank at most $r_0(X_0)$, because every such pair is contained in some deletion of X_2 .

By Lemma 2.6, X_0 has one of the types

$$Ax, \quad Ay, \quad BC, \quad By, \quad Cy.$$

We first rule out Ax . Suppose $X_0 = Ax$. Then $r_0(X_0) = 4$, so X_2 cannot contain an Ay -pair or a BC -pair. The remaining goods have types

$$A, B, B, C, C, y.$$

If X_2 contains the remaining A -good, then it cannot contain y , and it also cannot contain both a B -good and a C -good. Hence X_2 has size at most 3, contradiction. If X_2 does not contain the remaining A -good, then X_2 is chosen from B, B, C, C, y , and avoiding BC again gives size at most 3, contradiction. Hence $X_0 \neq Ax$.

We are left with four cases:

$$X_0 = Ay, \quad X_0 = BC, \quad X_0 = By, \quad X_0 = Cy.$$

Case 1: $X_0 = Ay$. The remaining goods have types

$$A, B, B, C, C, x.$$

Since $r_0(Ay) = 6$, agent 0's feasibility imposes no pair restriction on X_2 . The only restriction is that X_2 cannot contain an exceptional triple. Equivalently, X_2 cannot contain a B -good, a C -good, and one good of type A or x .

If X_2 contains both a B -good and a C -good, then it contains neither the A -good nor x . Thus

$$X_2 = BBCC \quad \text{and} \quad X_1 = Ax.$$

Deleting one B -good from X_2 leaves BCC , and

$$r_1(Ax) = 1 < 2 = r_1(BCC).$$

So agent 1 strongly envies agent 2.

If X_2 contains no B -good, then

$$X_2 = ACCx \quad \text{and} \quad X_1 = BB.$$

Deleting the A -good from X_2 leaves CCx , and

$$r_1(BB) = 1 < 4 = r_1(CCx).$$

Again agent 1 strongly envies agent 2.

Finally, if X_2 contains no C -good, then

$$X_2 = ABBx \quad \text{and} \quad X_1 = CC.$$

Deleting x from X_2 leaves ABB , and

$$r_1(CC) = 1 < 5 = r_1(ABB).$$

Thus agent 1 strongly envies agent 2 in every subcase.

Case 2: $X_0 = BC$. The remaining goods have types

$$A, A, B, C, x, y.$$

Since $r_0(BC) = 5$, the only forbidden pair in X_2 is Ay . In addition, X_2 cannot contain an exceptional triple.

First, $y \notin X_2$. Indeed, if $y \in X_2$, then X_2 contains no A -good. Since $|X_2| = 4$, this forces

$$X_2 = BCxy,$$

which contains the exceptional triple BCx , contradiction.

Thus X_2 is chosen from A, A, B, C, x . It cannot contain both the B -good and the C -good, because then, having size 4, it would also contain either an A -good or x , creating an exceptional triple. Hence the only possibilities are

$$X_2 = AACx, \quad X_1 = By,$$

or

$$X_2 = AABx, \quad X_1 = Cy.$$

In the first subcase, deleting one A -good from X_2 leaves ACx , and

$$r_1(By) = 3 < 4 = r_1(ACx).$$

In the second subcase, deleting one A -good from X_2 leaves ABx , and

$$r_1(Cy) = 6 < 7 = r_1(ABx).$$

Here ABx is exceptional for agent 1, since $\sigma(ABx) = BCx$. Thus agent 1 strongly envies agent 2 in both subcases.

Case 3: $X_0 = By$. The remaining goods have types

$$A, A, B, C, C, x.$$

Since $r_0(By) = 3$, X_2 cannot contain an Ax -pair or a BC -pair.

If $x \in X_2$, then X_2 contains no A -good. But then, to have size 4, it would have to contain the B -good and at least one C -good, creating a forbidden BC -pair. Hence $x \notin X_2$.

Now X_2 is chosen from A, A, B, C, C , still avoiding BC . It cannot contain the B -good, since otherwise it could contain no C -good and would have size at most 3. Therefore

$$X_2 = AACC \quad \text{and} \quad X_1 = Bx.$$

Deleting one A -good from X_2 leaves ACC , and

$$r_1(Bx) = 1 < 2 = r_1(ACC).$$

So agent 1 strongly envies agent 2.

Case 4: $X_0 = Cy$. The remaining goods have types

$$A, A, B, B, C, x.$$

Again $r_0(Cy) = 3$, so X_2 cannot contain an Ax -pair or a BC -pair.

If $x \in X_2$, then X_2 contains no A -good. But then, to have size 4, it would have to contain a B -good and the C -good, creating a forbidden BC -pair. Hence $x \notin X_2$.

Now X_2 is chosen from A, A, B, B, C , still avoiding BC . It cannot contain the C -good, since otherwise it could contain no B -good and would have size at most 3. Therefore

$$X_2 = AAB B \quad \text{and} \quad X_1 = Cx.$$

Deleting one A -good from X_2 leaves $AB B$, and

$$r_1(Cx) = 4 < 5 = r_1(AB B).$$

So agent 1 strongly envies agent 2.

In every possible case, agent 1 strongly envies agent 2, contradicting that X is EFX. Therefore no allocation of size pattern $(2, 2, 4)$ is EFX. $\square$

Allocations of type $(2, 3, 3)$

Proposition 2.8. *No allocation of size pattern $(2, 3, 3)$ is EFX.*

Proof. Suppose, for contradiction, that $X = (X_0, X_1, X_2)$ is an EFX allocation with $|X_0| = 2$ and $|X_1| = |X_2| = 3$. Since X is EFX, agent 0 is EFX-feasible. By Lemma 2.6, the five possible configurations for X_0 are

$$Ax, \quad BC, \quad Ay, \quad By, \quad Cy.$$

We examine these five cases. All rank comparisons below are read from Table 3; when a triple appears, its rank is the maximum rank of its internal pairs unless it is explicitly exceptional.

Case 1: $X_0 = Ax$. The remaining goods are of types

$$A, B, B, C, C, y.$$

Since $r_0(Ax) = 4$, no triple assigned to agent 1 or 2 can contain a BC -pair or an Ay -pair. Hence the split is either $AB B \mid CCy$ or $ACC \mid BBy$. In the first split, the holder of $AB B$ strongly envies the holder of CCy : for either $i \in \{1, 2\}$, deleting a C -good from CCy leaves a Cy -pair, and

$$r_i(AB B) < r_i(Cy).$$

In the second split, the holder of ACC strongly envies the holder of BBy : deleting a B -good from BBy leaves a By -pair, and

$$r_i(ACC) < r_i(By).$$

Therefore, no matter how we allocate, one agent strongly envies the other.

Case 2: $X_0 = BC$. The remaining goods have types

$$A, A, B, C, x, y.$$

Since $r_0(BC) = 5$, the only forbidden internal pair is Ay . Hence the triple containing y must contain no A type item. Thus one of the following splits occurs:

$$BCy \mid AAx, \quad Bxy \mid AAC, \quad Cxy \mid AAB.$$

In the case $BCy \mid AAx$, the holder of AAx strongly envies the holder of BCy , since deleting B from BCy leaves a Cy -pair and $r_i(AAx) < r_i(Cy)$. In the case $Bxy \mid AAC$, the holder of AAC strongly envies the holder of Bxy , since deleting x from Bxy leaves a By -pair and $r_i(AAC) < r_i(By)$. Finally, in the case $Cxy \mid AAB$, the holder of AAB strongly envies the holder of Cxy , since deleting x from Cxy leaves a Cy -pair and $r_i(AAB) < r_i(Cy)$. The needed rank comparisons for the two possible holders are

$$(r_1(AAx), r_2(AAx)) = (1, 1) < (6, 3) = (r_1(Cy), r_2(Cy)),$$

$$(r_1(AAC), r_2(AAC)) = (2, 5) < (3, 6) = (r_1(By), r_2(By)),$$

and

$$(r_1(AAB), r_2(AAB)) = (5, 2) < (6, 3) = (r_1(Cy), r_2(Cy)).$$

Case 3: $X_0 = Ay$. There is no further restriction from agent 0's feasibility condition, because $r_0(Ay) = 6$.

Let R be the triple containing the unique remaining A -good, and let S be the other triple. We show that the holder of S strongly envies the holder of R .

First suppose S is held by agent 1. Since S contains no A -good, all its goods are of type B, C, x . Under agent 1's relabelling, these types become C, A, x , so S has rank at most 4. If R contains an AB -pair, then

$$r_1(AB) = r_0(BC) = 5,$$

and agent 1 strongly envies the holder of R . If R contains no AB -pair, then, since R contains A , it must be either ACC or ACx . If $R = ACC$, then $S = BBx$, and deleting one C -good from R leaves an AC -pair with

$$r_1(BBx) < r_1(AC).$$

If $R = ACx$, then $S = BBC$, and deleting the A -good from R leaves a Cx -pair with

$$r_1(BBC) < r_1(Cx).$$

Thus agent 1 strongly envies the holder of R in all cases.

The argument for agent 2 is symmetric. If S is held by agent 2, then $r_2(S) \leq 4$. If R contains an AC -pair, then

$$r_2(AC) = r_0(BC) = 5,$$

so agent 2 strongly envies the holder of R . If R contains no AC -pair, then R is either ABB or ABx . If $R = ABB$, then $S = CCx$, and deleting one B -good from R leaves an AB -pair with

$$r_2(CCx) < r_2(AB).$$

If $R = ABx$, then $S = BCC$, and deleting the A -good from R leaves a Bx -pair with

$$r_2(BCC) < r_2(Bx).$$

Again, agent 2 strongly envies the holder of R . Therefore strong envy is unavoidable when X_0 has type Ay .

Case 4: $X_0 = By$. The remaining goods have types

$$A, A, B, C, C, x.$$

Since $r_0(By) = 3$, no triple can contain an Ax -pair or a BC -pair. Thus x cannot be placed with an A -good. If x were placed with the single remaining B -good, then its third partner would have to be a C -good, creating a forbidden BC -pair. Hence x must be placed with the two C -goods. The split is forced:

$$CCx \mid AAB.$$

The holder of CCx strongly envies the holder of AAB : after deleting one A -good from AAB , the remaining pair is AB . If the holder is agent 1, then

$$r_1(CCx) < r_1(AB),$$

and if the holder is agent 2, then

$$r_2(CCx) < r_2(AB).$$

Case 5: $X_0 = Cy$. The remaining goods have types

$$A, A, B, B, C, x.$$

Again $r_0(Cy) = 3$, so no triple can contain an Ax -pair or a BC -pair. Thus x cannot be placed with an A -good. If x were placed with the single C -good, then its third partner would have to be a B -good, creating a forbidden BC -pair. Hence x must be placed with the two B -goods. The split is forced:

$$BBx \mid AAC.$$

The holder of BBx strongly envies the holder of AAC : after deleting one A -good from AAC , the remaining pair is AC . If the holder is agent 1, then

$$r_1(BBx) < r_1(AC),$$

and if the holder is agent 2, then

$$r_2(BBx) < r_2(AC).$$

In every possible case, one of agents 1 and 2 strongly envies the other, contradicting that X is EFX. Therefore no allocation of size pattern $(2, 3, 3)$ is EFX. $\square$

Proof of Theorem 2.1. By Lemma 2.4, it suffices to rule out one cyclic relabelling of each possible size pattern.

Let $X = (X_0, X_1, X_2)$ be any allocation. If some bundle has size at most one, cyclically relabel so that this bundle is in position 0. This case is ruled out by Proposition 2.5. We may therefore assume that every bundle has size at least two. Since there are eight goods and three agents, the size multiset is either $\{2, 2, 4\}$ or $\{2, 3, 3\}$. In the first case, cyclically relabel to obtain ordered size pattern $(2, 2, 4)$, which is ruled out by Proposition 2.7. In the second case, cyclically relabel so that the unique pair is in position 0, giving ordered size pattern $(2, 3, 3)$, which is ruled out by Proposition 2.8. Hence no allocation satisfies EFX. $\square$

3 Weak Preferences to Submodular and Subadditive Valuations

In this section, we realise the ordinal obstruction by explicit cardinal valuations. We first record a transfer lemma: any cardinal profile that preserves the strict part of the ordinal preferences inherits the ordinal non-existence result.

Lemma 3.1. *Let $\{\succeq_i\}_{i \in N}$ be a weak preference profile over bundles for which no allocation satisfies EFX. Suppose that, for each agent i , the valuation function $v_i : 2^M \rightarrow \mathbb{R}$ is consistent with the strict part of $\succeq_i$, in the sense that*

$$S \succ_i T \implies v_i(S) > v_i(T)$$

for all bundles $S, T \subseteq M$. Then no allocation satisfies EFX under the cardinal valuation profile $(v_i)_{i \in N}$.

Proof. Consider any allocation $X = (X_1, \dots, X_n)$. Since X is not EFX with respect to the weak preference profile, there exist agents $i, j \in N$ and a good $g \in X_j$ such that

$$X_j \setminus \{g\} \succ_i X_i.$$

By consistency of v_i with the strict preference order, this implies

$$v_i(X_j \setminus \{g\}) > v_i(X_i).$$

Thus, agent i envies agent j even after the removal of g , so X is not EFX under the cardinal valuations. $\square$

3.1 Subadditive Valuations: An Approximation Barrier

Let $\lambda = 2^{-1/6}$. We define valuation functions from the rank functions r_0, r_1, r_2 constructed in Section 2. For each agent i , define

$$v_i(S) = \begin{cases} 0, & \text{if } S = \emptyset, \\ \lambda^{7-r_i(S)}, & \text{if } S \neq \emptyset. \end{cases}$$

Proposition 3.2. *The profile (v_0, v_1, v_2) is normalised, monotone, and subadditive.*

Proof. Normalisation is immediate. Monotonicity follows from Observation 2.3, since $\lambda < 1$.

It remains to prove subadditivity. Fix an agent i and let $S, T \subseteq M$. If either set is empty, the claim is immediate. Otherwise, both sets are non-empty, and hence

$$v_i(S) \geq \lambda^6 = \frac{1}{2}, \quad v_i(T) \geq \lambda^6 = \frac{1}{2}.$$

Since the value of every set is at most 1, we obtain

$$v_i(S) + v_i(T) \geq 1 \geq v_i(S \cup T).$$

Thus v_i is subadditive. $\square$

Lemma 3.3 (One rank gap gives the α -gap). *Let $\alpha > \lambda$. If $S, T \neq \emptyset$ and*

$$r_i(T) > r_i(S),$$

then

$$v_i(S) < \alpha v_i(T).$$

Proof. Since the ranks are integer-valued, $r_i(T) > r_i(S)$ implies $r_i(T) \geq r_i(S) + 1$. Equivalently, $7 - r_i(S) \geq 7 - r_i(T) + 1$. By the definition of the valuation function, and using $0 < \lambda < 1$, we therefore have

$$v_i(S) = \lambda^{7-r_i(S)} \leq \lambda^{7-r_i(T)+1} = \lambda \lambda^{7-r_i(T)} = \lambda v_i(T).$$

Since $\alpha > \lambda$, it follows that $v_i(S) \leq \lambda v_i(T) < \alpha v_i(T)$ as desired. $\square$

Theorem 3.4. *For every $\alpha \in (2^{-1/6}, 1]$, the subadditive valuation profile (v_0, v_1, v_2) admits no α -EFX allocation.*

Proof. Suppose, for a contradiction, that X is an α -EFX allocation. Since the underlying ordinal profile has no EFX allocation by Theorem 2.1, there exist agents i, j and a good $g \in X_j$ such that

$$r_i(X_j \setminus \{g\}) > r_i(X_i).$$

If $X_i = \emptyset$, then $v_i(X_i) = 0$, while $X_j \setminus \{g\}$ is non-empty and hence has a positive value, so the α -EFX inequality is violated.

Otherwise, both X_i and $X_j \setminus \{g\}$ are non-empty. By Lemma 3.3, we obtain

$$v_i(X_i) < \alpha v_i(X_j \setminus \{g\}),$$

which again contradicts α -EFX. Therefore no α -EFX allocation exists for the valuation profile. $\square$

Together with Proposition 3.2, this proves Theorem 1.2.

3.2 Submodular Valuations via Weighted Coverage

Instance Description

The weighted-coverage valuation below is chosen to realise the strict comparisons of the rank order from Section 2. It depends only on which of the five types A, B, C, x, y are represented in a bundle. Thus, after the formula is given, strict consistency can be checked over the 32 possible type supports rather than over all 2^8 bundles.

For a set $R \subseteq M$, define the coverage indicator

$$\chi_R(S) = \mathbb{1}_{\{S \cap R \neq \emptyset\}}.$$

We define the base valuation $u_0 : 2^M \rightarrow \mathbb{R}_{\geq 0}$ by

$$\begin{aligned} u_0(S) = & \chi_B(S) + \chi_C(S) \\ & + 3\chi_{AU\{x\}}(S) + 3\chi_{BU\{y\}}(S) + 3\chi_{CU\{y\}}(S) \\ & + 8\chi_{AUB}(S) + 8\chi_{AUC}(S) \\ & + 9\chi_{BU\{x,y\}}(S) + 9\chi_{CU\{x,y\}}(S) \\ & + 2\chi_{AUBU\{y\}}(S) + 2\chi_{AUCU\{y\}}(S). \end{aligned}$$

The other two valuations are cyclic relabellings:

$$u_1(S) = u_0(\sigma(S)), \quad u_2(S) = u_0(\sigma^2(S)).$$

Proposition 3.5. *The profile (u_0, u_1, u_2) consists of weighted coverage valuations. In particular, the valuations are normalised, monotone, and submodular.*

Proof. The function u_0 is a non-negative linear combination of coverage indicators. Equivalently, for each displayed set R , one may introduce a coverage atom whose weight is the coefficient of χ_R . Thus u_0 is a weighted coverage valuation, and hence is normalised, monotone, and submodular.

The valuations u_1 and u_2 are obtained from u_0 by relabelling the goods, and therefore inherit the same properties. $\square$

Proof of Consistency

We now show that the weighted coverage function u_0 is consistent with the weak monotone-order from Section 2.

For a bundle S , write $\text{supp}(S) \subseteq \{A, B, C, x, y\}$ for the set of types represented in S . For example,

$$\text{supp}(AABB) = AB, \quad \text{supp}(AACx) = ACx.$$

In the next table, words such as AB , ACx , and $BCxy$ denote type supports, not type multisets.

Lemma 3.6 (Support collapse). *For every bundle $S \subseteq M$, both $r_0(S)$ and $u_0(S)$ depend only on $\text{supp}(S)$.*

Proof. This is immediate for u_0 , since each term in its definition is a coverage indicator depending only on which types are represented in S .

For r_0 , repeating a type does not create a higher rank: all singletons and same-type pairs have rank 1. A non-exceptional triple inherits the largest rank of an internal pair, while an exceptional triple exists exactly when the support contains ABC or BCx . For sets of size at least four, r_0 is the maximum rank of an internal triple. Thus $r_0(S)$ is determined entirely by $\text{supp}(S)$. $\square$

The only remaining point is that the coverage values strictly refine the rank order. Since both r_0 and u_0 depend only on type support, it is enough to check the 32 possible supports listed in Table 5. Thus the table is the finite certificate for strict consistency, not merely an illustration of the valuation.

Table 5: Exhaustive support table for r_0 and u_0 . Words denote type supports. Thus the row AB , for example, includes $AB, AAB, ABB, AABB$.

$r_0(S)$	possible values of $u_0(S)$	type supports $\text{supp}(S)$
0	0	$\emptyset$
1	21, 23, 28, 31, 35	$x, A, B, C, y, xy, Bx, Cx$
2	36	AB, AC
3	37, 40	By, Cy, Bxy, Cxy
4	41, 45	Ax, ABx, ACx
5	46	BC, BCy
6	47, 48	$Ay, Axy, ABx, ACy, ABxy, ACxy$
7	49	$ABC, ABCx, ABCy, ABCxy, BCx, BCxy$

Lemma 3.7 (Strict consistency of the weighted-coverage realization). *For all bundles $S, T \subseteq M$,*

$$r_0(S) > r_0(T) \implies u_0(S) > u_0(T).$$

Consequently, for every agent $i \in \{0, 1, 2\}$,

$$r_i(S) > r_i(T) \implies u_i(S) > u_i(T).$$

Proof. By Lemma 3.6, it suffices to check type supports. There are exactly $2^5 = 32$ type supports, since each of A, B, C, x, y is either present or absent. The supports in Table 5 are exhaustive: the table lists the empty support and

$$8 + 2 + 4 + 3 + 2 + 6 + 6 = 31$$

non-empty supports.

The value ranges in successive rank rows are strictly separated:

$$0 < 21 \leq 35 < 36 < 37 \leq 40 < 41 \leq 45 < 46 < 47 \leq 48 < 49.$$

Equivalently, the minimum value in each positive-rank row is larger than the maximum value in the preceding row. Therefore $r_0(S) > r_0(T)$ implies $u_0(S) > u_0(T)$. The statement for agents 1 and 2 follows from

$$r_i(S) = r_0(\sigma^i(S)), \quad u_i(S) = u_0(\sigma^i(S)).$$

□

Theorem 3.8. *The weighted-coverage valuation profile (u_0, u_1, u_2) admits no EFX allocation.*

Proof. By Proposition 3.5, the profile consists of weighted coverage valuations, and by Lemma 3.7, it preserves every strict comparison of the ordinal profile constructed in Section 2. Since this ordinal profile admits no EFX allocation by Theorem 2.1, Lemma 3.1 implies that the weighted coverage profile also admits no EFX allocation. □

As an independent check, we also ran the EFX-verification code of Akrami et al. [2026] on the valuation profiles constructed above. For each of the three profiles, namely the ordinal profile, the subadditive profile, and the weighted coverage profile, the code found no EFX allocation. These computations are not used in the proof and serve only as an additional verification of the explicit examples.

4 Discussion and Open Directions

Beyond the counterexamples themselves, the construction gives a useful way to organize the search for structured failures of EFX. The first step is ordinal: we specify only how agents rank bundles, and the non-existence proof uses only those rankings. The typed structure of the goods and the cyclic symmetry between the agents keep the proof small enough to check by hand: the case analysis is over bundle types such as Ay , $BBCC$, and $CCx \mid AAB$, rather than over arbitrary valuation tables.

The cardinal constructions then answer a separate question: which valuation classes can express the same ordinal pattern? For the subadditive result, the ranks are converted into separated numerical levels. For the submodular result, the weighted-coverage formula is chosen so that its values strictly preserve the relevant rank order. Thus the subadditive and weighted-coverage examples share the same underlying preference pattern, even though the cardinal valuations used to realize it are quite different.

This suggests a general strategy for finding counterexamples in restricted valuation classes. One can first look for monotone ordinal profiles with no EFX allocation and enough structure to

be understandable. One can then ask whether those profiles, or small modifications of them, can be realized by valuations from a target class. The present paper shows that this approach can reach both monotone subadditive valuations and weighted coverage valuations.

The relabeling symmetry is also worth studying in its own right. In our examples, all agents share a single valuation template: they differ only in the permutation of the goods through which they evaluate bundles. This makes the profile far more restricted than an arbitrary heterogeneous one, yet still expressive enough to produce instances which admit no EFX allocations for both subadditive and weighted-coverage valuations. It would be interesting to understand which positive EFX results survive under this identical-up-to-relabeling assumption, and how useful this symmetry class is as a testing ground for fair division conjectures.

For subadditive valuations, the present construction leaves a large gap. A $1/2$ -EFX allocation is known to exist, while our instance rules out α -EFX for every $\alpha \in (2^{-1/6}, 1]$. Improving the upper bound may require more goods or agents, but it may also require a different obstruction, or a realization in which the bad comparisons depend on more detailed cardinal information rather than only on ordinal rank gaps.

The weighted-coverage valuations are strict subclass of submodular valuations. A natural next question is whether the failure persists in other well structured submodular classes. A gross-substitutes counterexample would already be significant, and an OXS counterexample would be stronger still. These classes are highly structured and economically important, but they are not contained in the MMS-feasible class. The known three-agent EFX existence result for two arbitrary monotone valuations and one MMS-feasible valuation therefore does not rule them out.

Another direction is to look for genuinely four-agent obstructions, where the failure uses a four-way interaction rather than extending a three-agent counterexample by adding agents or goods. Such examples might reveal new structural phenomena or lead to stronger approximation barriers.

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